

Flow-Based Market Coupling

TET4185

Spring 2023

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1 Introduction

This memo aims to provide a high-level overview of the flow-based (FB) methodology. FB market coupling is the preferred market design in the Network Code on Capacity Calculation and Congestion Management (NC CACM). The basic idea separating FB from the current Coordinated Net Transmission Capacity (CNTC) is introducing a simplified grid model in the market clearing algorithm.

With CNTC, the market algorithm considers only commercial exchanges between bidding zones. Actual physical flows are left to the TSO to manage. Congestions are solved on a "border by border" basis by regulating the net positions on each side of the congested border. In order to manage the actual physical flows effectively, the TSOs have to prioritise and allocate capacity to certain borders to manage the effect of transit flows and internal congestion. As transit flows are hard to predict, capacity calculation in a meshed grid becomes more complex.

The scope of FB is different from CNTC. The transmission capacities provided to the market come together with information on the physical flows (linearised as such) on all Critical Network Elements (CNE) induced by a change in the net position in every bidding zone¹. The flows are then monitored, and overloads are managed directly by the market algorithm. The TSO can provide maximum capacity to the market, and the market algorithm will find the optimal welfare economic flow on all grid components by itself.

The material presented here is, to a significant extent, based on "Methodology and concepts for the Nordic Flow-Based Market Coupling Approach", a report on Nordic Flow-based Market Coupling.

2 General Principles of Flow-based Allocation

Based on the laws of physics in the interconnected transmission network, any commercial exchange between adjacent regional power markets results in a physical power flow partly on the direct inter-regional transmission lines and partly across all available parallel paths in the network. The power flow on those parallel paths, also known as loop flow, is ignored in CNTC and might cause network security problems in the control area of each TSO.

The flow-based allocation (FBA) mechanism aims to facilitate cross-border trades by providing maximum available inter-regional transmission capacity without compromising the system security. More specifically, the FBA mechanism describes the interdependency of cross-border commercial transactions between regional power markets and the (physical) power

¹ Bidding Zones may differ from price areas in that one price area may contain several bidding zones

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flow those transactions cause on all cross-border interconnections in the FBA region. In case of transmission congestion, a commercial exchange between any two power markets within that region has an effect on all congested interconnectors. Besides, this allocation procedure also accounts for the netting of the induced power flow on every single interconnector between neighbouring control areas by evaluating the impact of all cross-border transactions on that given transmission link. Ultimately, this yields more efficient utilisation of interregional cross-border capacities.

Since the TSO does not have to prioritise capacity on the certain border in advance, more solutions are available to the market algorithm. This implies that theoretically, the solution domain (See Section 2.2), given to the market by a FB capacity allocation, is as large as or larger than the CNTC domain. All CNTC market solutions are available to the FBMC, but the FBMC provides access to solutions outside the CNTC solution domain. Whenever the optimal solution is within the CNTC domain, both market designs will find it, but the FBMC may find an optimum outside what is available to the CNTC. In theory, the FBMC has to be more efficient than the CNTC at the same level of system security, while practical implementation may sometimes prove otherwise.

2.1 Grid constraints limit the domain for the market solution

The generic market optimisation problem may be formulated as:

CNTC formulation:	Objective function:	Maximise welfare economic surplus
	Subject to:	$\sum \text{NPs} = 0,$ CNTC constraints

FB formulation:	Objective function:	Maximise welfare economic surplus
	Subject to:	$\sum \text{NPs} = 0,$ FB constraints

NP (Net Positions) = Supply - Demand

Without diving into the mathematics of these relations, they point to the fact that the objective function is the same for CNTC and FB, whereas the constraints are different. The objective function is to maximise total welfare economic surplus in the power market, which is the sum of producer surplus, consumer surplus and congestion income.

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The constraints limit the solution domain to the market optimisation problem, and they are the key to understanding why FB may provide a better solution than CNTC. The fact is that, given the same level of operational security, the boundaries of the FB domain will always be located on or outside the boundaries of the CNTC domain. This implies that if the optimum market solution is found within the CNTC domain, both CNTC and FB will find the same solution. However, the optimum solution may be within the FB domain but outside the CNTC domain.

2.2 CNTC and FB constraints

We can illustrate the difference between the FB and the CNTC using the simple three-node grid in Figure 1. In this example, all lines have a thermal capacity of 1000MW and equal impedance (equal "electrical distance"). Node C is a consumption node, and nodes A and B are generation nodes. The question faced by the TSO is how much capacity can be provided to the market for each line (CNE).

At the time of capacity calculation (D-1)², the Transmission System Operator (TSO) does not know which area (A or B) is to produce. The physical property of the grid is, however, known. Due to the described grid topology, one MW produced in A will induce a flow of 2/3 MW on line AC, 1/3 MW on line AB, and 1/3 MW on line BC. The same holds for generation B, of which -1/3 appears on AB, 1/3 on AC and 2/3 on BC. These sensitivity factors are commonly referred to as Power Transfer Distribution Factors (PTDF). Node C is deemed to be the slack node, and all power injected in A, and B is absorbed in this slack. The same holds for node C itself: all power injected in C is absorbed in the slack node C, as such, node C individually has no influence on the flows in the grid. The flow influence of each node to each line comprises the PTDF matrix, which is shown in Figure 1.

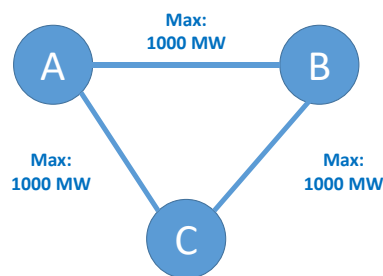


Figure 1. Grid with three nodes

² The capacity calculation starts at D-2. Final values are provided to the market at D-1

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Three bidding zones connected by three lines with equal impedances (equal "electrical distance"). In this example, the three lines are the constraining CNEs. N-1 contingencies are ignored

The PTDF factors, translating the change in net positions into physical line flows, are not provided to the market algorithm under CNTC. Within the CNTC approach, transit flows are ignored, and the algorithm only relates to the total provided capacity on each border. Figure 1 implies that one MW produced in A and consumed in C will bring a "CNTC load" of, e.g., 0.5 MW on the lines AB and BC and 0,5 MW on AC. Bluntly setting the CNTCs to thermal limit values would allow the CNTC market algorithm to carry 2000 MW of trade from A to C, or from B to C, even though it is not physically feasible and will create an overload.

Table 1. PTDF matrix of the grid

Line	RAM ³	A	B	C
A->B	1000	1/3	-1/3	0
A->C	1000	2/3	1/3	0
B->C	1000	1/3	2/3	0

The physical reality in our simple three-node example, reflected by the PTDFs, is that 2000 MW generated in A will load AC with $2/3 \cdot 2000 = 1333$ MW, which is above the thermal capacity of that line. The maximum generation (net position) in each of A and B that is possible in order to avoid overloads is 1500 MW (however, not simultaneously). The operator (TSO) has to limit the total export from A and B to this level under the CNTC approach. One possible set of CNTC capacities that can be provided to the market will be a capacity of 750 MW on AC, BC and AB, which gives a secure CNTC solution domain. This is also the maximum simultaneous net positions that can be obtained within the CNTC approach in the network illustrated in Figure 1. The solution CNTC domain is illustrated in Figure 2.

The solution domain indicates which net positions are physically safe within a particular grid topology. (This is elaborated further in Section 5.1). The CNTC market optimisation boils down to finding the optimum market position inside the (CNTC) domain (indicated by the blue lines in Figure 2).

When FB constraints are provided to the market, the solution domain (or security domain) will change. The FB constraints consist of information on flows induced (PTDFs) and the CNE-capacities given to the market called Remaining Available Margins (RAM). In our small three-node example, the RAM is 1000 MW on each line (N-1 and security margins are ignored). As

³ Remaining Available Margin

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with CNTC, a net position of 1500 MW for each A and B is still feasible within FB. However, the larger maximum simultaneous net position of 1000 MW for A and B also becomes possible with FB. This corresponds to the net positions of A=1000, B=1000, and C=-2000 (point 1 in Figure 2). Flow induced on AB is $1000 \cdot (2/3) + 1000 \cdot (1/3) = 1000$, the flow on BC is $1000 \cdot (1/3) + 1000 \cdot (2/3) = 1000$, and the flow on AB is $1000 \cdot (1/3) + 1000 \cdot (-1/3) = 0$.

Another market position accessible in FB but not in CNTC is a net position of 2000 MW for both A and B (but not simultaneously), Illustrated as point 2 in Figure 2. A net position of 2000 MW for A corresponds to the following net positions A=2000, B=-1000, C=-1000, and the flow-induced on line AB is $2000 \cdot 1/3 - 1000 \cdot (-1/3) - 1000 \cdot 0 = 1000$.

If all such "extra" points are added to the former CNTC domain, we have the FB domain, which is shown in grey in Figure 2. In all situations where the optimal solution is found within the grey area but outside the blue area, the FB solution is a better solution in terms of welfare economics than the CNTC solution.

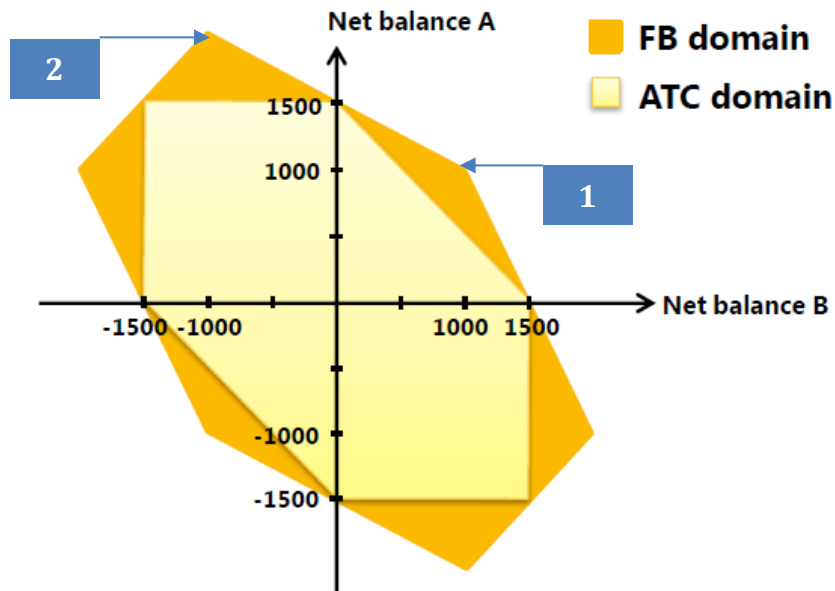


Figure 2. CNTC and FB domains (The figure shows a safe CNTC and FB domain derived from the grid in Figure 1 with the same level of system security)

All points on the FB boundaries reflect congested situations somewhere in the grid that will induce price differences in all nodes without implying that all lines are congested simultaneously. However, these market positions are not possible in CNTC because the CNTC algorithm does not know the actual physical flows (the PTDFs) between bidding zones.

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2.3 Capacity allocation at the power exchange (Euphemia)

The common market algorithm used in Northwestern Europe is called Euphemia. In principle, Euphemia is doing what is described in the generic model in Section 2.1 above. Euphemia is calculating the welfare maximum, but several "optimum" price/volume combinations may exist due to the stepwise nature of real-world bids. Euphemia uses a heuristic to choose a solution when steps are overlapping. This implies that the interpretation of shadow prices in Euphemia might not necessarily be in line with theory, which requires smooth continuous curves.

2.4 Overview of the Flow-Based Market coupling process

Three distinct phases can be identified in the operational procedure in a power market irrespective of whether it is based on a FB or a CNTC methodology; pre-market coupling, market coupling and post-market coupling.

Pre-market coupling is the preparatory phase where the TSOs do the capacity calculation. This is a TSO- only operation where the solution domain for the market is calculated. The pre-market coupling basically consists of:

- 1) Creating a "base case" containing the expected grid topology for the next day together with expected net positions of all bidding zones and corresponding flows on all CNEs.
- 2) Defining Generation Shift Keys (GSKs), CNEs, the corresponding outages and Remedial Actions
- 3) Defining and applying operational experience (Final Adjustment value, FAV) in order to adjust the FB domain.
- 4) Applying GSKs and CNEs to do the parameter calculation creating PTDFs and market margins (capacities).
- 5) Verification of the FB parameters, making sure that the domain is proper.
- 6) Sending the parameters to the power exchange.

The pre-market coupling starts in the evening on D-2 and lasts until 10:00 on D-1 when the parameters are published on the power exchange website.

Market coupling is the actual solving of the market. The power exchange carries out this phase. The Market coupling process consists of:

- 1) Publishing the parameters to the market
- 2) Collecting bids from the market players
- 3) Calculate the market equilibrium
- 4) Publishing the market result

The market coupling process starts at 10:30 and ends at 13:00 when market results are published.

The **post-market coupling** process is carried out at the TSOs. This process consists of:

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- 1) Verification of the market results
- 2) Congestion income sharing
- 3) Analyses of operational security

2.5 Uncertainty

The reliability margin is the fundamental element in managing uncertainty in the capacity calculation (Flow Reliability Margin or FRM in FB). Due to uncertainty, the power system operator cannot precisely predict what flow will be realised on each CNE in the hour of operation. The flow may be larger or smaller than anticipated, and if the flow turns out larger, there may be an overload on a CNE. In order to reduce the probability of physical overloads to an acceptable risk level, some of the capacity of a CNE will be retained from the market as an FRM. The capacity provided to the market will be the maximum capacity on a CNE, less the FRM. The size of the FRM will typically be based on a statistical evaluation of the deviations between the flows estimated by the FB method and the actual flows observed.

There are many reasons why uncertainty occurs in the capacity calculation, such as temperature, precipitation, fuel prices and sun. However, two uncertainties are fundamental and specific to FB procedure. The first is the linearisation of the grid model. The second is the manner in which we choose to aggregate the node-to-CNE PTDFs to zone-to-CNE PTDFs. On the other hand, the flows between bidding zones are solved by the market itself when FB is applied. This means that transit flows are calculated in the FBMC, reducing this uncertainty.

All uncertainties will, however in the end, be reflected in the Flow Reliability Margin (FRM), which reduces the market capacity compared to the physical capacity

3 Network code requirements related to FB

Several Network Codes (NCs) are at the moment being prepared for the European Power Market. The NCs are the EU-wide implementation of the Third Framework Program for the European Power Market, and they will define a common set of rules for the European electricity markets. The NCs are still under development, so the final text and content are not yet known. But the basic framework is there, and most issues have been covered.

Several NCs are being developed that influence the FB methodology. The Network Code on Capacity Allocation and Congestion Management (NC CACM) is the most relevant one in this perspective.

3.1 Network Code on Capacity Allocation and Congestion Management (NC CACM)

From a FB point of view, the most important NC is the NC CACM which will have implications for capacity calculation and allocation and also sets forth requirements for the input to these processes. Two issues are of particular interest to FB:

- 1) Requirements for the capacity calculation methodology

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2) Requirements for a European Common Grid Model (methodology)

CACM specifies that the methodology for capacity calculation should either be a coordinated NTC (CNTC) or FB approach for both the day-ahead and intraday timeframes. The preferred approach, however, shall be a FB capacity calculation unless:

- a) In regions where interdependencies between cross-zonal capacities are low, and
- b) There is no added value to applying the FB approach

Furthermore, the CACM states that each Coordinated Capacity Calculator applying the FB Approach shall use the Common Grid Model (CGM), Generation Shift Keys (GSKs) and contingencies to calculate the Power Transfer Distribution Factors (PTDFs).

The CGM itself shall represent the European interconnected system and facilitate transmission capacities to be calculated in a coordinated manner. The CGM will include the transmission system with the locations of generation and load units relevant for capacity calculation, grid topology, and rules to change this during capacity calculation.

In further detail, the CGM shall contain descriptions of the specific scenarios to be used, where each scenario is to be defined per market segment for both the day-ahead and the intraday market. The code prescribes how to merge individual (national) grid models into a CGM, in which each individual/national grid model shall be based on the best forecast for the operational conditions in the given scenario at the time the model is produced. All TSOs are to ensure that the individual grid models contain all information necessary to run a load flow analysis, and in appropriate instances, the models should be prepared to run dynamic analyses.

4 Pre-market coupling

In this Section, we explain all activities the TSOs performed to produce grid constraints to be used by the allocation mechanism (i.e. capacity calculation) in detail. We dive into how the FB parameters are calculated and which simplifications are made. As such, this Section dives into the details of what was explained on a high level in Section 2.1.

We start off by explaining the relation between the physical grid (described by the AC load flow equations) and the PTDFs that the TSOs provide to the market algorithm. Subsequently, we explain the other central parameter, the market margin (Remaining Available Margin, RAM). The rest of the Section is dedicated to explaining how to deal with GSKs, CNEs, DC cables, reliability margins, and further details behind the FB parameters.

4.1 *The sensitivity parameters (PTDF)*

The power transfer distribution factors (PTDFs) in FB are calculated based on a standard set of power flow AC equations. This Section starts with a short introduction to the basics of the AC power flow equations and how the DC power flow representation of the grid used in FB is derived.

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4.1.1 Power flow equations

The active and reactive power flows in a steady state can be described by the following non-linear⁴ equations:

$$P_i = V_i \sum_{k=1}^N V_k (G_{ik} \cos(\delta_i - \delta_k) + B_{ik} \sin(\delta_i - \delta_k)) \quad (1)$$

$$Q_i = V_i \sum_{k=1}^N V_k (G_{ik} \sin(\delta_i - \delta_k) - B_{ik} \cos(\delta_i - \delta_k)) \quad (2)$$

Where,

P_i	Active power balance in node i (per unit MW)
Q_i	Reactive power balance in node i (per unit Mvar)
i, k	Node number
N	Number of nodes
V_i	Voltage magnitude in node i
δ_i	Voltage angle of node i
δ_k	Voltage angle of node k
G_{ik}	Conductance between node i and k with negative sign
G_{ii}	Sum of all conductances connected to node i
B_{ik}	Susceptance between node i and k with negative sign
B_{ii}	Sum of all susceptances connected to node i

The two active and reactive node-balance equations above show the balance of each node in the AC network as the sum of the flow on branches and shunts connected to the node. These power flow equations aim to determine all buses' voltages (magnitude and angle). If the voltages are known, it is possible to determine the power flows, losses and currents.

The terms conductance (G) and susceptance (B) above constitute the complex admittance (measured in Siemens) given by Y :

⁴ The equations are nonlinear due to the trigonometric sine and cosine elements

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$$Y = \frac{I}{V} = G + jB = \left(G = \frac{R}{R^2 + X^2}, B = -\frac{X}{R^2 + X^2} \right) = \frac{R}{R^2 + X^2} - j \frac{X}{R^2 + X^2} \quad (3)$$

Here r represents the resistance and x the reactance of the connected circuit. The circuit can be e.g. a line or transformer. Figure 3 below illustrates the notation used in the equations.

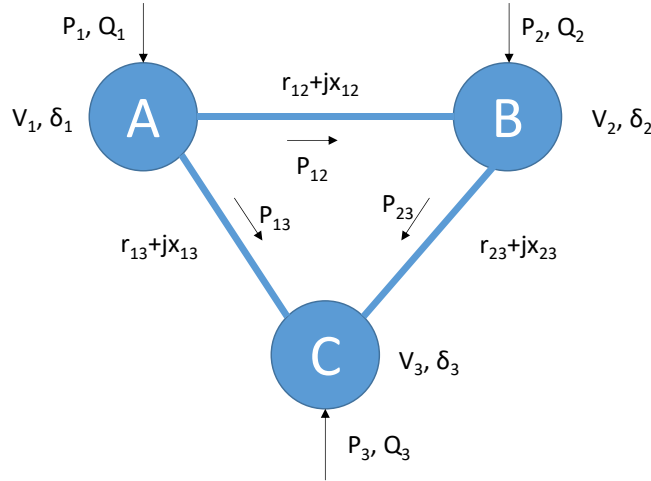


Figure 3. Example grid with three nodes

The node and line parameters used in the power flow equations are illustrated in Figure 3.

The FB market solution for the energy market in Europe is based on a linearisation of equations (1) and (2). The following is a brief description of the approximations made in linearisation.

The first approximation is based on the fact that the grid resistance is typically much less than the grid reactance. If r is small compared to x , it can be observed in (3) that:

$$r \ll x \rightarrow G \approx 0 \text{ and } B \approx -\frac{1}{x} \quad (4)$$

The power flow equations (1) and (2) now become:

$$P_i = V_i \sum_{k=1}^N V_k (B_{ik} \sin(\delta_i - \delta_k)) \quad (5)$$

$$Q_i = V_i \sum_{k=1}^N V_k (-B_{ik} \cos(\delta_i - \delta_k)) \quad (6)$$

The second approximation is based on the fact that the voltage angle between two buses is typically small during the stable operation of the power system. Sine and cosine can be approximated as:

$$\sin \delta_i \approx \delta_i \text{ and } \cos \delta_i \approx 1 \quad (7)$$

The power flow equations (5) and (6) now become:

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$$P_i = V_i \sum_{k=1}^N V_k (B_{ik} (\delta_i - \delta_k)) \quad (8)$$

$$Q_i = V_i \sum_{k=1}^N V_k (-B_{ik}) \quad (9)$$

The third approximation is based on the assumption that all the voltage magnitudes will be almost equal to the reference voltage. If one uses the per unit (pu) system, where the reference voltage is normalised to 1.0, the following approximation can be made:

$$V_i, V_j \approx 1 \quad (10)$$

The power flow equations (8) and (9) now become

$$P_i = \sum_{k=1}^N B_{ik} (\delta_i - \delta_k) \quad (11)$$

$$Q_i = \sum_{k=1}^N -B_{ik} \quad (12)$$

Since Q_i turns into a constant term, it has no impact on the flow in the network, and only the equation of active power, P_i is needed to compute the voltage angles for a given level of consumption and production. This method is labelled "DC power flow".

The equations so far have established the relationship between the power injection in each node and the flows of the connected lines. The voltage angles, in an example with three nodes, can be computed from the following equation:

$$[\delta] = \begin{bmatrix} P_1 \\ P_2 \\ P_3 \end{bmatrix} = \begin{bmatrix} B_{12} + B_{13} & -B_{12} & -B_{13} \\ -B_{21} & B_{21} + B_{23} & -B_{23} \\ -B_{31} & -B_{32} & B_{31} + B_{32} \end{bmatrix} \begin{bmatrix} \delta_1 \\ \delta_2 \\ \delta_3 \end{bmatrix} = [B][\delta] = [Ybus][\delta] \quad (13)$$

In equation (13), the $Ybus$ -matrix is symmetric as $B_{12}=B_{21}$ and corresponds to the negative susceptance between nodes 1 and 2. The diagonal elements are equal to the sum of all susceptances connected to the node. This $Ybus$ -matrix is also called the B -matrix or the bus admittance matrix. Notice that δ is the vector with the voltage angle at the nodes.

The voltage angles are then given by Equation (14).

$$[\delta] = \begin{bmatrix} \delta_1 \\ \delta_2 \\ \delta_3 \end{bmatrix} = \begin{bmatrix} B_{12} + B_{13} & -B_{12} & -B_{13} \\ -B_{21} & B_{21} + B_{23} & -B_{23} \\ -B_{31} & -B_{32} & B_{31} + B_{32} \end{bmatrix}^{-1} \begin{bmatrix} P_1 \\ P_2 \\ P_3 \end{bmatrix} = [Zbus][P] \quad (14)$$

Where, the inverted $Ybus$ matrix is referred to as the $Zbus$ or bus impedance matrix.

There is an infinite set of possible solutions of equation (14), since the injected power P only depends on the differences between the voltage angle δ , and not on the absolute values. The

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problem is that there is a dependency in the three equations, implying that one equation may be obtained from the other two. As such, the inverse of the Y_{bus} -matrix is singular. In order to have a unique solution, at least one of the diagonal elements must contain an additional value, creating a "reference point" for the absolute values of the system. This node is referred to as the slack-bus or slack-node and is set to $\delta = 0$. If node 1 is set as the slack-node, $\delta_1 = 0$, the corresponding column and row (column 1 and row 1) in the Y_{bus} matrix in (13) are eliminated. The remaining voltage angles can then be calculated by

$$\begin{bmatrix} \delta_2 \\ \delta_3 \end{bmatrix} = \begin{bmatrix} B_{21} + B_{23} & -B_{32} \\ -B_{32} & B_{31} + B_{32} \end{bmatrix}^{-1} \begin{bmatrix} P_2 \\ P_3 \end{bmatrix} \quad (15)$$

Given the voltage angles in each node, the active power flow between two nodes can be calculated with equation (16).

$$P_{ik} = B_{ik}(\delta_i - \delta_k) \quad (16)$$

Instead of eliminating a row and a column in the Y_{bus} -matrix for the slack-bus, as in (15), another method is to add "+1" to one of the diagonal elements. Hence, if node 1 is selected as the slack node, the voltage angles can be calculated as

$$\begin{bmatrix} \delta_1 \\ \delta_2 \\ \delta_3 \end{bmatrix} = \begin{bmatrix} 1 + B_{12} + B_{13} & -B_{12} & -B_{13} \\ -B_{21} & B_{21} + B_{23} & -B_{23} \\ -B_{31} & -B_{32} & B_{31} + B_{32} \end{bmatrix}^{-1} \begin{bmatrix} P_1 \\ P_2 \\ P_3 \end{bmatrix} = [Z_{bus}][P] \quad (17)$$

The final result is the same in approach in (15) and (17), but the latter might be more straightforward.

4.1.2 Calculating the PTDF

The power transfer distribution factor (PTDF) reveals a certain node's flow participation on a certain branch. For example, the PTDF for node 1 at line 1-2 in the example above is denoted $PTDF_{12,1}$. The PTDF matrix then includes each node's PTDF for all lines in the system.

The PTDF is derived in the following way. For example, assume that additional power, ΔP_i , is fed into the network in node 1. From (17), the change in angles in node 1 and 2, given this injection, can be calculated as

$$\Delta \delta_1 = \Delta P_1 (1 + B_{12} + B_{13}) = \Delta P_1 (Z_{bus_{11}}) \quad (18)$$

$$\Delta \delta_2 = \Delta P_1 (-B_{21}) = \Delta P_1 (Z_{bus_{21}}) \quad (19)$$

The elements multiplied with P_2 and P_3 in (17) do not change the angles since the power is injected in node 1 in this example, leaving ΔP_2 and ΔP_3 to be zero.

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The change of active power flow in line 1-2 due to this injection in node 1 can now be calculated by combining (16) with (18) and (19).

$$\Delta P_{12} = B_{12} (\Delta \delta_1 - \Delta \delta_2) = B_{12} (Zbus_{11} - Zbus_{21}) \Delta P_1 \quad (20)$$

If ΔP_1 is set to unity (1.0) the PTDF for node 1 at line 1-2 become

$$PTDF_{12,1} = \Delta P_{12} = B_{12} (\Delta \delta_1 - \Delta \delta_2) = B_{12} (Zbus_{11} - Zbus_{21}) \quad (21)$$

More generic, the PTDF can be expressed as

$$PTDF_{ik,n} = B_{ik} (Zbus_{in} - Zbus_{kn}) \quad (22)$$

Equation (22) should be interpreted as the PDTF value from node n to the line between nodes i and k . By repeating this procedure for all nodes and all lines, the PTDF matrix can be computed. The matrix describes how the net balance of the nodes influences the power transfers on the lines. Remember here that B_{ik} is the susceptance of the line between node i and k , which corresponds to the inverse of the reactance of the line since we assume that the resistance is negligible.

4.2 Example: Calculating PTDFs

Figure 4 below shows a three-node network where the nodal transfer PTDFs are going to be calculated. The impedances of the lines are included in the figure, being the sum of resistance and reactance. The slack node is located in node 3 in this example.

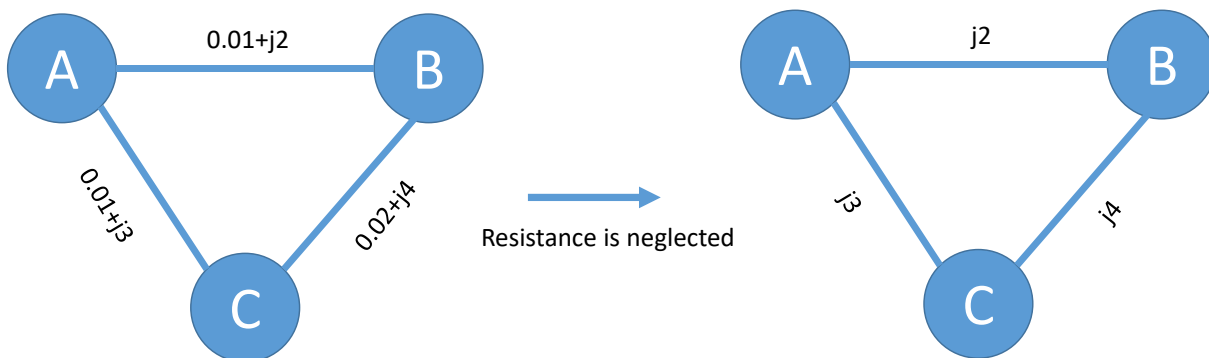


Figure 4 Example grid with three nodes

The node and line parameters used in the power flow equations are illustrated in the figure. The line resistance is negligible compared to the reactance (e.g. line 1-2 has a $2/0.01=200$ times

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higher reactance); hence, the assumption in (4) can be applied. The DC power flow approximation is applied.

The Y_{bus} matrix, given by (13), is defined by the data in Figure 4. Recall that the susceptance between two nodes equals the inverse of the reactance for the line, since the resistance was neglected.

$$Y_{bus} = \begin{bmatrix} \frac{1}{2} + \frac{1}{3} & -\frac{1}{2} & -\frac{1}{3} \\ -\frac{1}{2} & \frac{1}{2} + \frac{1}{4} & -\frac{1}{4} \\ -\frac{1}{3} & -\frac{1}{4} & \frac{1}{3} + \frac{1}{4} \end{bmatrix} \quad (23)$$

As in (17), the Z_{bus} matrix is then constructed by adding "+1" to the diagonal element corresponding to the slack node in the bus matrix in (23), followed by an inverse operation. Node 3 is, in this example, selected as a slack node.

$$Z_{bus} = \begin{bmatrix} \frac{1}{2} + \frac{1}{3} & -\frac{1}{2} & -\frac{1}{3} \\ -\frac{1}{2} & \frac{1}{2} + \frac{1}{4} & -\frac{1}{4} \\ -\frac{1}{3} & -\frac{1}{4} & \frac{1}{3} + \frac{1}{4} + 1 \end{bmatrix}^{-1} = \begin{bmatrix} 3.00 & 2.33 & 1.00 \\ 2.33 & 3.22 & 1.00 \\ 1.00 & 1.00 & 1.00 \end{bmatrix} \quad (24)$$

The PTDF value from node n for the line between nodes i and k can then be calculated with (22).

$$PTDF_{ik,n} = B_{ik} (Z_{bus_{in}} - Z_{bus_{kn}}) \quad (25)$$

For example, the PTDF value from node 1 to the line between node 1 and 2 can be calculated as Equation 26

$$PTDF_{12,1} = B_{12} (Z_{bus_{11}} - Z_{bus_{21}}) = \left(\frac{1}{2}\right) (3.00 - 2.33) = 0.33 = 33\% \quad (26)$$

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For production in node 1, 33% of the power will flow on line 1 to 2. For consumption (which is the negative production), the effect will be the reverse, i.e. the line is loaded in the opposite direction.

For each line ik (row) and node n (column), the $PTDF_{ik,n}$ is calculated, resulting in the following PTDF matrix (nodal transfer PTDF matrix, to be precise) with node 3 being the slack-node:

$$PTDF = \begin{matrix} & \begin{matrix} \text{Node} \\ 1 & 2 & 3 \end{matrix} \\ \begin{matrix} 1-2 \\ 1-3 \\ 2-3 \end{matrix} & \begin{bmatrix} 0.33 & -0.44 & 0 \\ 0.67 & 0.44 & 0 \\ 0.33 & 0.57 & 0 \end{bmatrix} \end{matrix} \quad (27)$$

4.3 Remaining Available Margin (RAM)

The PTDF matrix is one of two fundamental parameters for providing grid constraints to market optimisation at the PX. The other is the Remaining Available Margin (RAM). This is the "free margin" that the allocation mechanism on CNEs can use. The RAM differs from the max capacity of the CNE as in the following equation:

$$RAM = F_{\max} - FRM - FAV - Fref' \quad (28)$$

Where,

RAM	The Remaining Available Margin
$F_{\max}$	The maximum allowed flow on the CNE
$F_{ref'}$	The reference flow at zero net positions when using the computed PTDF
FRM	The Flow Reliability Margin
FAV	The Final Adjustment Value

In this equation, $F_{ref'}$ is the reference flow at zero net position that is obtained by using the calculated PTDF matrix from the base case:

$$Fref' = Fref - PTDF \times NP^{BC} \quad (29)$$

Where,

$Fref$ The loading of the CNEs in the base case given the net positions reflected in the base case

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NP^{BC} Net position of all bidding zones in the base case

The relation between the net position, flow and RAM is illustrated in Figure 4. The RAMs on CNEs with their associated PTDF factors form the so-called FB constraints:

$$PTDF \times NP \leq RAM \quad (30)$$

In general, the RAMs will be positive, and the flows on the CNEs, induced by the net positions that are optimised by the market coupling mechanism, will be restricted by those values. However, a certain CNE may be pre-congested (already congested before the actual allocation). In this case, the F_{ref}' exceeds the $F_{max}-FRM-FAV$ value resulting in a negative RAM. In the case that a negative RAM is provided to the market coupling algorithm, the market is enforced to relieve that congestion irrespective of the market preferences. In the example below, the constraint enforces the induced flow on the CNE to be -10 or smaller (e.g. -15).

Example of negative RAM:

$$PTDF_a \times NP(A) + PTDF_b \times NP(B) + PTDF_c \times NP(C) \leq -10 \quad (31)$$

The market coupling mechanism will find the most efficient way to relieve congestion.

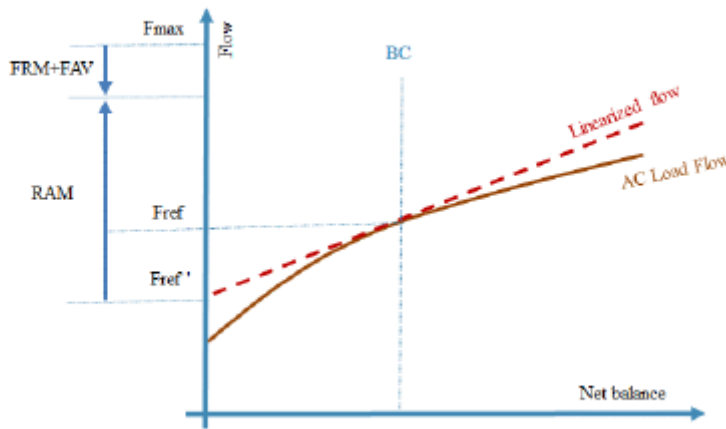


Figure 4. Relation between flow, net position and RAM

5 FB constraints

The RAMs on the CNEs with their associated PTDF factors form the so-called flow-based constraints. Note that a RAM of a CNE is defined in one direction only. If both directions are to be monitored, two CNEs have to be defined, one for each direction. The FB constraints represent the grid constraints in the Market Coupling mechanism:

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Objective function: Maximise economic surplus (Welfare)
 Subject to: $\sum \text{NPs} = 0$
 FB constraints

5.1 FB domain and its indicators

The FB constraints will ultimately limit the market to a fixed set of possible solutions. The TSO calculates these constraints in what is known as "capacity calculation". The capacity calculation is thereby crucial from a welfare economic point of view. The methodology and principles of FB are discussed in the following Section based on a three-node network⁵ illustrated in Figure 5.

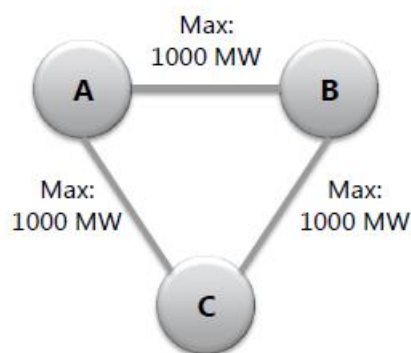


Figure 5. Example: three bidding areas

The network has the following characteristics:

- Equal impedances (equal "electrical distance")
- Max flow on the branches: 1000 MW
- With zero NPs, the lines between the Bidding Zones are unloaded

⁵ Example based on Schavemaker Guest lecture on Flow-Based Concept and Methodology,

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The maximum export from A to any of the two other bidding areas amounts to 1500 MW. This is due to the physical reality where the electricity always follows the path of least resistance and flows fan out following Kirchhoff's laws. In the example, this implies that the flow from A to C will distribute as 1/3'd on the longest route and 2/3'ds on the shortest route as in Figure 10.

The FB constraints represent a simplified linearised grid model. The model reflects both the impact (flow) of import/export from every bidding zone on each CNEs in the grid, and the total capacity available for the market on each CNE. This is illustrated in the graphs and table in Figures 10 and 11, where A, B and C are bidding zones, and the connecting tie lines are the CNEs.

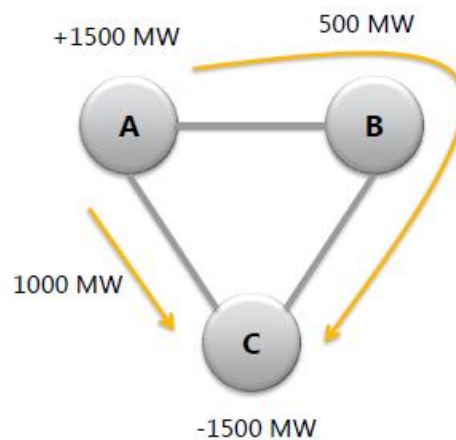
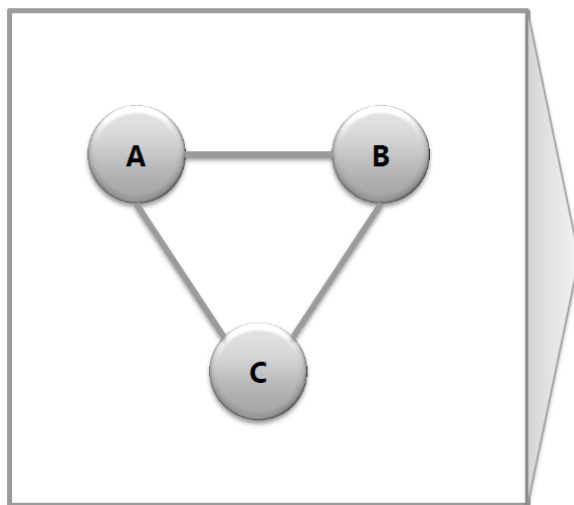


Figure 6. Physical flows in a three-node network. Equal impedance on each CNE/line

The impacts of the various bidding zones on the CNEs are evaluated in the form of PTDFs, being a sensitivity factor stating how much (%) of one MW of export from a certain bidding zone will occur on a particular CNE. Considering the PTDFs in both directions provides us with the complete set of FB parameters, as illustrated in the table in Figure 7.

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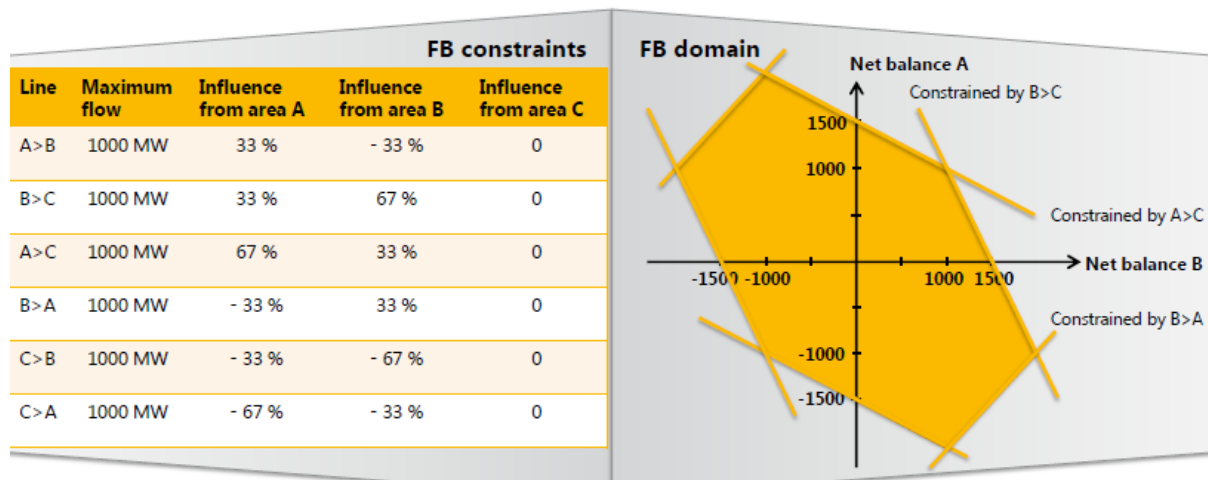


FB constraints ('grid model'):

Line	Margins		PTDF factors		
	Maximum flow	Influence from area A	Influence from area B	Influence from area C	
A>B	1000 MW	33 %	- 33 %	0	
B>C	1000 MW	33 %	67 %	0	
A>C	1000 MW	67 %	33 %	0	
B>A	1000 MW	- 33 %	33 %	0	
C>B	1000 MW	- 33 %	- 67 %	0	
C>A	1000 MW	- 67 %	- 33 %	0	

Figure 7. FB parameters in a three- node network, max flows and PTDFs, PTDFs are calculated for both directions

The FB constraints for the three-node system define the FB domain: the import/export positions available for the market solution while not jeopardising the grid security. The domain is visualised in

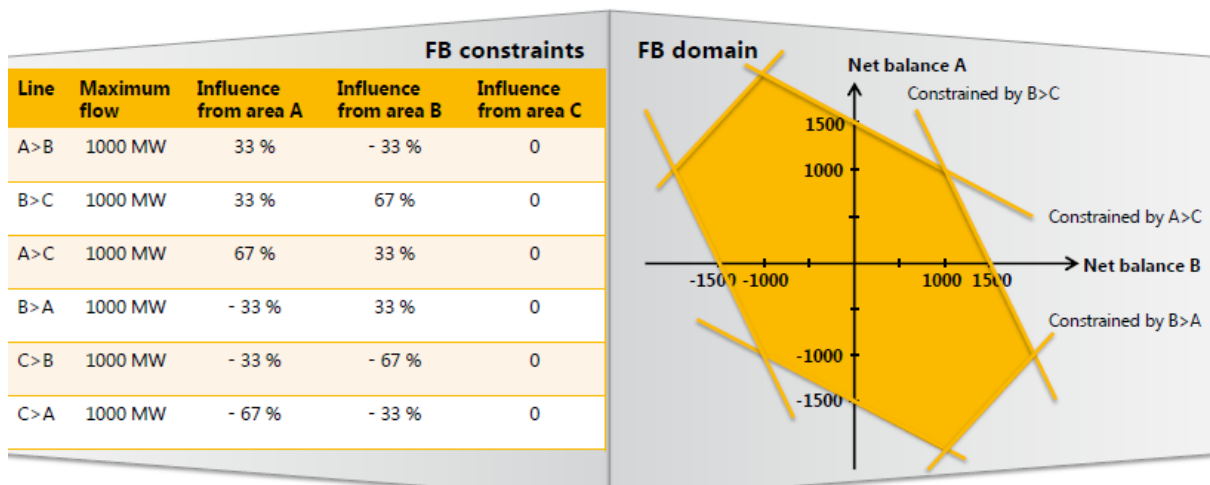


Figure

8.

The lines limiting the FB domain can be easily found from the FB parameters. Let's zoom in on the constraint imposed by line A>C. When the NP(A) = 0, NP(B) can reach a value of $1000/0.33 = 3000$ MW (i.e. the intersection point at the x-axis). When the NP(B) = 0, NP(A) can reach a value of $1000/0.67 = 1500$ MW (i.e. the intersection point at the y-axis). With these two points, the line ("Constrained by AC") can be drawn in

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Figure

8.

We can continue to find the next constraint, imposed by line B>C. When NP(A) is zero, NP(B) can reach the value of $1000/0.67 = 1500$ MW. When NP(B) is at zero, NP(A) can reach a value of $1000/0.33 = 3000$ MW. This is the line "Constrained by B>C".

in
Figure

8, we see that both points NP(A)=3000 MW, and NP(B)=3000 MW are invalid points outside the FB domain at all times. Both points NP(A)=1500 MW and NP(B)=1500 MW are valid points, but they can't occur at the same time though, e.g. (NP(A), NP(B), NP(C)) = (1500, 0, - 1500) is a point that is within the FB domain. The point NP(A)=1000 MW and NP(B)=1000 MW is also valid point, and they can occur at the same time: (NP(A), NP(B), NP(C)) = (1000,1000,-2000).

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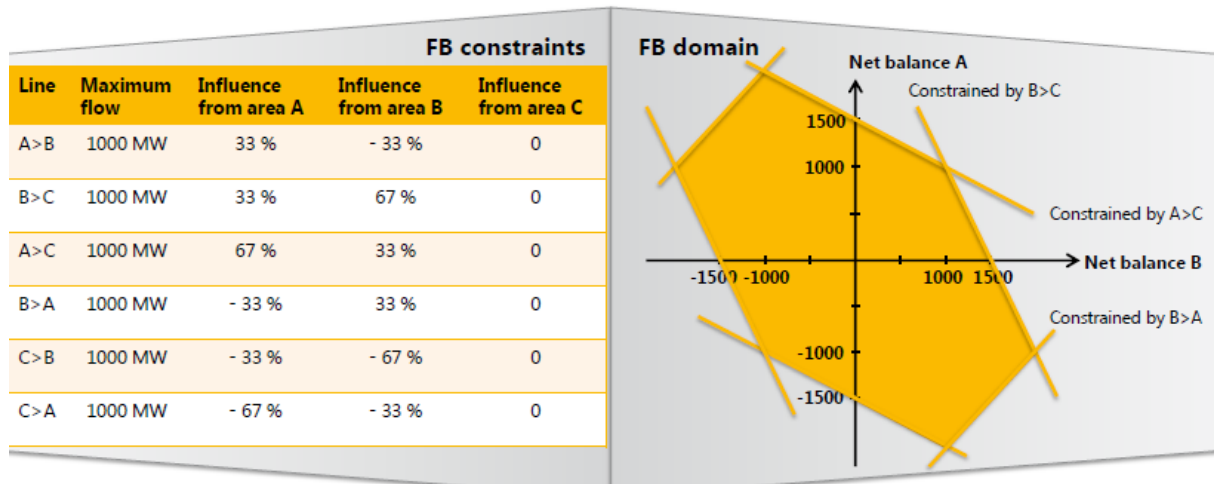


Figure 8. FB domain

The FB domain also provides additional means to characterise the FB constraints:

- The vertices of the FB domain, in the example above, the points (1000,1000,-2000) and (1500,0,1500) are examples of vertices. The complete set of vertices gives the domain, as illustrated in Figure 12
- The maximum net positions feasible in the FB domain, in the example above: max NP(A) = 2000, min NP(A) = -2000, max NP(B) = 2000, min NP(B) = -2000
- The maximum bilateral exchanges feasible within the FB domain (by assuming that no other exchanges take place), e.g. the maximum exchange from A to C equals 1500 MW in our example
- The volume of the FB domain

5.2 Comparison to a CNTC domain

The indicators that can be extracted from the FB domain can also be derived for the CNTC domain and provide a basis for comparison:

- The vertices of the FB and CNTC domain
- The maximum net positions feasible in the FB and CNTC domain
- The maximum bilateral exchanges feasible within the FB and CNTC domain
- The volume of the FB and CNTC domain
 - The volume of the common part of the domains (intersection)
 - The volume of the FB domain not in the CNTC domain
 - The volume of the CNTC domain not in the FB domain

The CNTC and the FB domains for the three-node example are shown in Figure 9. We can see that the CNTC domain is entirely inside the FB domain. Indeed, more options are available to the market in the FB setup than in a CNTC setup.

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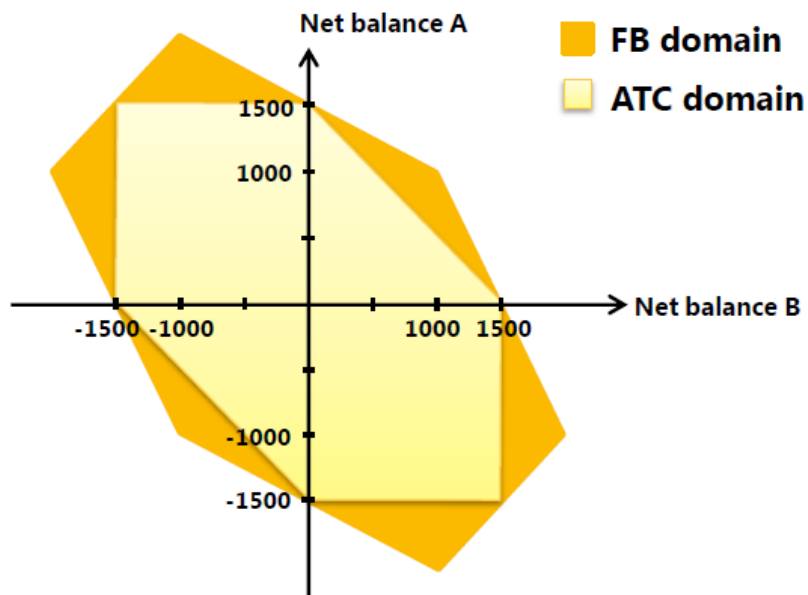


Figure 9. Comparing CNTC and FB solution domains. The domain is derived from Figure 7, and the CNTC values are all set to 750 MW